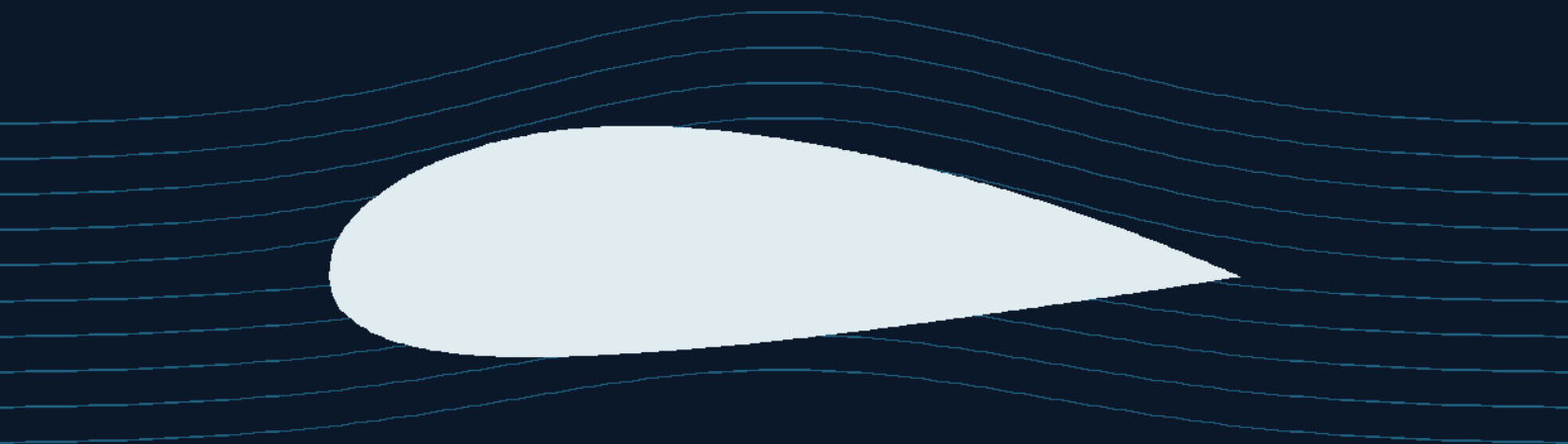


CFD AERODYNAMIC STUDY

NACA 2412 AIRFOIL

Steady RANS analysis, mesh sensitivity,
lift curve, drag polar and stall onset



ANSYS Fluent 2026 R1 Student

$U_{\infty} = 30 \text{ m/s}$ | $Re \sim 2.05e6$ | k-omega SST

Executive summary

This report documents a two-dimensional CFD investigation of a NACA 2412 airfoil at a freestream velocity of 30 m/s. The work was developed from a verified NACA 0012 baseline and then extended to the cambered NACA 2412, with particular attention to near-wall resolution, numerical convergence, aerodynamic coefficient normalization and angle-of-attack boundary conditions. The final sweep covers 0 to 18 degrees and captures the linear lift regime, the maximum-efficiency region, and the onset of stall.

At zero incidence, the CFD solution gives $C_l = 0.2174$ and $C_d = 0.0100$. Thin-airfoil theory predicts C_l approximately 0.228 from a zero-lift angle of about -2.08 degrees, corresponding to a difference of approximately 4.6%. The maximum lift measured in the discrete sweep is $C_{l,max} = 1.5241$ at 16 degrees. At 17 and 18 degrees the lift decreases while drag rises sharply, indicating that the solution has passed the lift peak. The highest measured aerodynamic efficiency is $C_l/C_d = 52.28$ at 8 degrees.

1. Objective and aerodynamic context

The NACA 2412 is a four-digit cambered airfoil with a 2% maximum camber located at 40% chord and 12% maximum thickness. Unlike the symmetric NACA 0012, it produces positive lift at zero geometric incidence. The study therefore provides a useful bridge between basic CFD verification and a more realistic aerodynamic polar analysis.

The main objectives were to obtain a numerically stable viscous RANS solution, verify the zero-incidence lift against thin-airfoil theory, construct $C_l(\alpha)$ and $C_d(\alpha)$, identify the best measured aerodynamic efficiency, and document the development of separation near the lift peak using velocity path lines.

2. Geometry, domain and mesh

The airfoil chord is $c = 1$ m. The NACA 2412 coordinates were generated analytically using the standard four-digit camber-line construction and a closed trailing-edge thickness polynomial. The two-dimensional computational domain extends approximately 5 chords upstream, 11 chords downstream and 2.5 chords above and below the chord line. The airfoil is removed from the fluid surface to create a single external-flow domain.

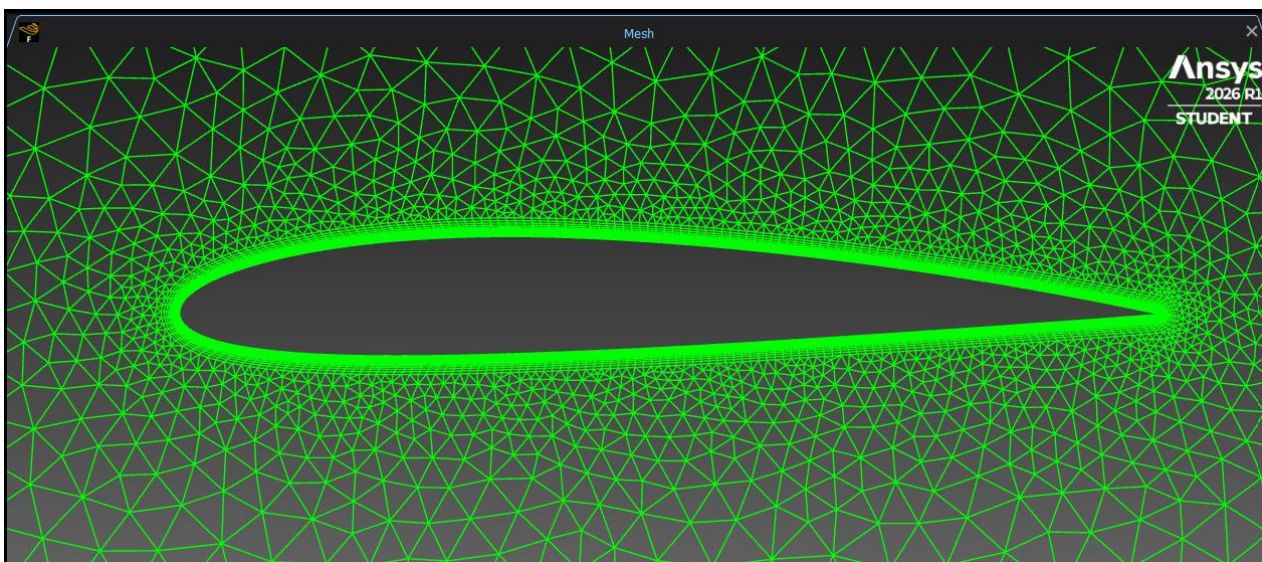


Figure 1 - Near-airfoil mesh and inflation region.

Near-wall resolution is provided by an explicit inflation layer on the airfoil. The first-layer height is 2.4×10^{-5} m (0.024 mm), with 30 layers and a growth rate of 1.18. The final NACA 2412 calculations used the refined mesh settings adopted after the baseline sensitivity study: global target size 0.15 m and airfoil edge size 0.005 m. The outer field is predominantly unstructured, while the inflation region resolves the boundary layer with highly stretched near-wall cells.

The initial NACA 2412 mesh quality assessment revealed a very small minimum orthogonal quality and a maximum skewness close to unity, associated with the sharp closed trailing-edge cusp. Rather than changing geometry, solver stabilization and a refined mesh were used. This is an important limitation: the trailing-edge cell quality should be improved in a production-grade study, preferably with a structured C-grid/O-grid or a controlled trailing-edge topology.

3. Physical and numerical model

Parameter	Value
Solver	Pressure-based, steady, 2D, double precision
Freestream velocity	30 m/s
Density	1.225 kg/m^3
Dynamic viscosity	$1.79 \times 10^{-5} \text{ Pa s}$
Chord Reynolds number	$\sim 2.05 \times 10^6$
Turbulence model	k-omega SST
Inlet turbulence	1% intensity; viscosity ratio 10
Pressure-velocity coupling	SIMPLE
Spatial discretization	Second Order Upwind (final converged runs)
Reference chord / area	1 m / 1 m^2 per unit span

The k-omega SST model was selected because it combines robust near-wall behavior with improved treatment of adverse pressure gradients compared with simpler eddy-viscosity closures. The final calculations were performed with second-order upwind discretization after using more conservative first-order settings and reduced under-relaxation during the stabilization phase.

A critical normalization check was the reference velocity used for force coefficients. Fluent initially retained $V_{\text{ref}} = 1 \text{ m/s}$, which inflated reported coefficients by a factor proportional to $1/V_{\text{ref}}^2$. Correcting V_{ref} to 30 m/s restored physically meaningful nondimensional coefficients without changing the underlying flow solution.

4. Angle-of-attack implementation

The airfoil geometry was kept fixed, and incidence was imposed by rotating the freestream vector. In the final Fluent setup, the convention that reproduced increasing positive lift used U_x velocity component proportional to the cosine of the sweep angle and U_y velocity component proportional to the sine of the sweep angle. The left and bottom boundaries were velocity inlets, while the top and right boundaries were pressure outlets. This arrangement is consistent with the direction of the imposed flow in the adopted domain orientation and was verified by the monotonic lift response from 0 to 8 degrees. Lift and drag were reported in axes aligned with the freestream.

alpha [deg]	Ux [m/s]	Uy [m/s]	Drag vector (x,y)	Lift vector (x,y)
0	30.00	0.00	(1.0000, 0.0000)	(-0.0000, 1.0000)
4	29.93	2.09	(0.9976, 0.0698)	(-0.0698, 0.9976)
8	29.71	4.18	(0.9903, 0.1392)	(-0.1392, 0.9903)
12	29.34	6.24	(0.9781, 0.2079)	(-0.2079, 0.9781)
14	29.11	7.26	(0.9703, 0.2419)	(-0.2419, 0.9703)
16	28.84	8.27	(0.9613, 0.2756)	(-0.2756, 0.9613)
17	28.69	8.77	(0.9563, 0.2924)	(-0.2924, 0.9563)
18	28.53	9.27	(0.9511, 0.3090)	(-0.3090, 0.9511)

5. Convergence and zero-incidence validation

Convergence was assessed using both residual behavior and force-coefficient stability. For C_l and C_d monitors, a stop criterion of $1e-4$ over a 20-iteration window was used after an initial ignored interval of 50 iterations. Near stall, convergence required substantially more iterations; the 18-degree case required a very high number of iterations before the reported forces stabilized. This sensitivity is consistent with the increasingly separated flow and is a warning that a steady RANS solution near stall should be interpreted as a mean steady approximation rather than definitive unsteady physics.

For a 0 degree sweep angle, the thin airfoil theory gives us a value of coefficient lift $C_l = 0.228$. The CFD value is 0.2174, a relative difference of about 4.6%. Given that thin-airfoil theory is inviscid and assumes a thin profile while the CFD model is viscous RANS, the agreement is strong and supports the consistency of the baseline solution.

6. Aerodynamic sweep results

alpha [deg]	C_l	C_d	C_l/C_d
0	0.2174	0.01000	21.74
4	0.6176	0.01250	49.41
8	1.0246	0.01960	52.28
12	1.3415	0.03340	40.16
14	1.4651	0.04410	33.22
16	1.5241	0.06050	25.19
17	1.4888	0.07480	19.90
18	1.4175	0.09606	14.76

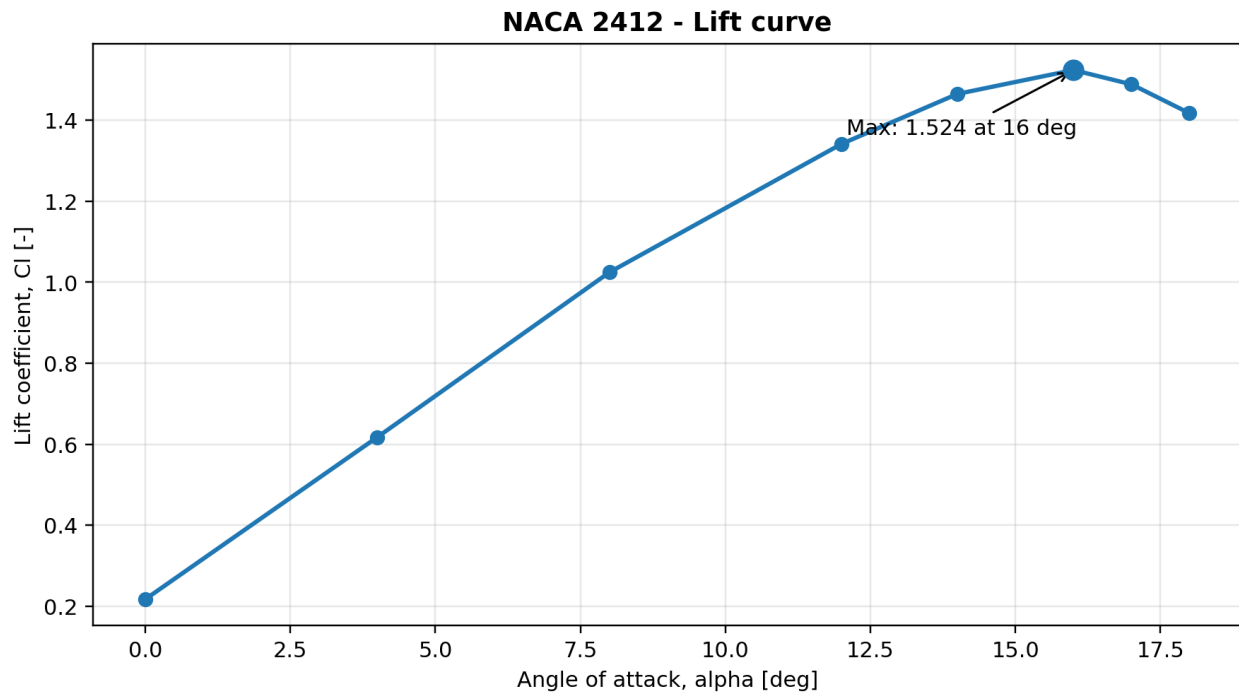


Figure 2 - Lift coefficient as a function of angle of attack.

The lift curve is nearly linear from 0 to 8 degrees. Over this interval, the measured slope changes, close to the classical two-dimensional thin-airfoil slope of about 0.110 per degree. Above 8 degrees, the curve progressively bends, indicating increasing nonlinear effects and separation. The highest lift calculated is $C_l = 1.5241$ at 16 degrees; C_l then falls to 1.4888 at 17 degrees and 1.4175 at 18 degrees.

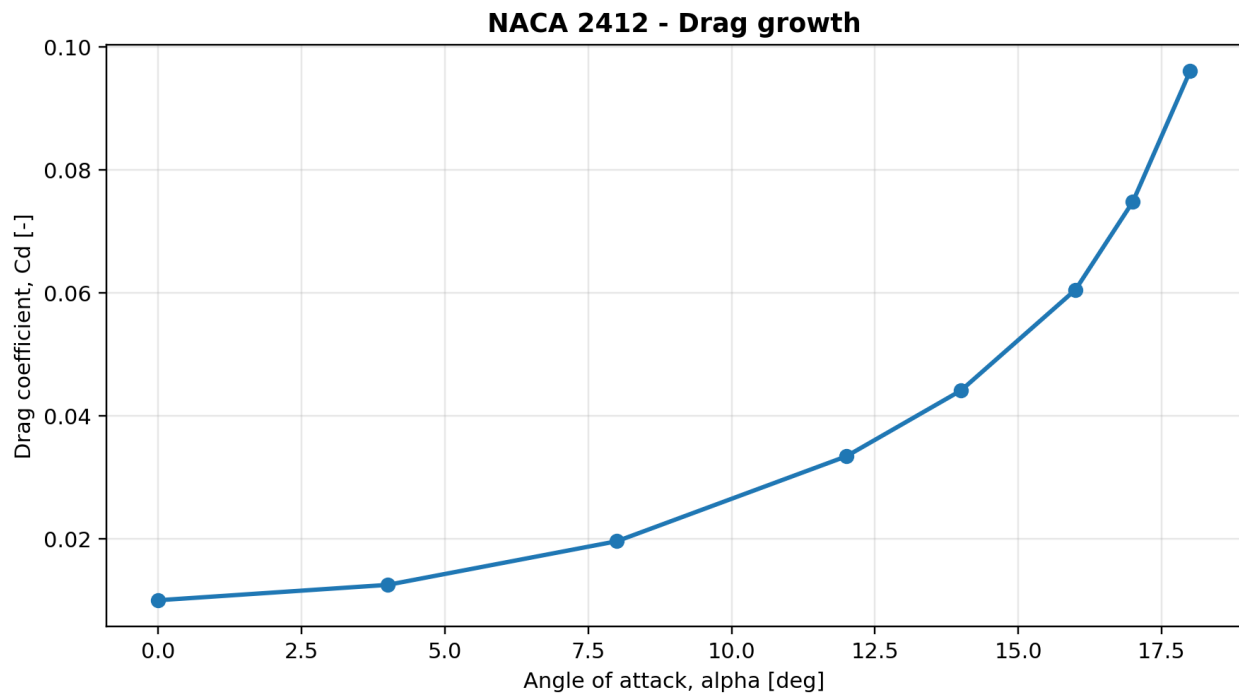


Figure 3 - Drag coefficient growth with angle of attack.

Drag increases modestly at low incidence and then rises rapidly as the flow approaches stall. Between 16 and 18 degrees, C_d increases from 0.0605 to 0.0961 while C_l decreases, a characteristic signature of post-peak aerodynamic behavior.

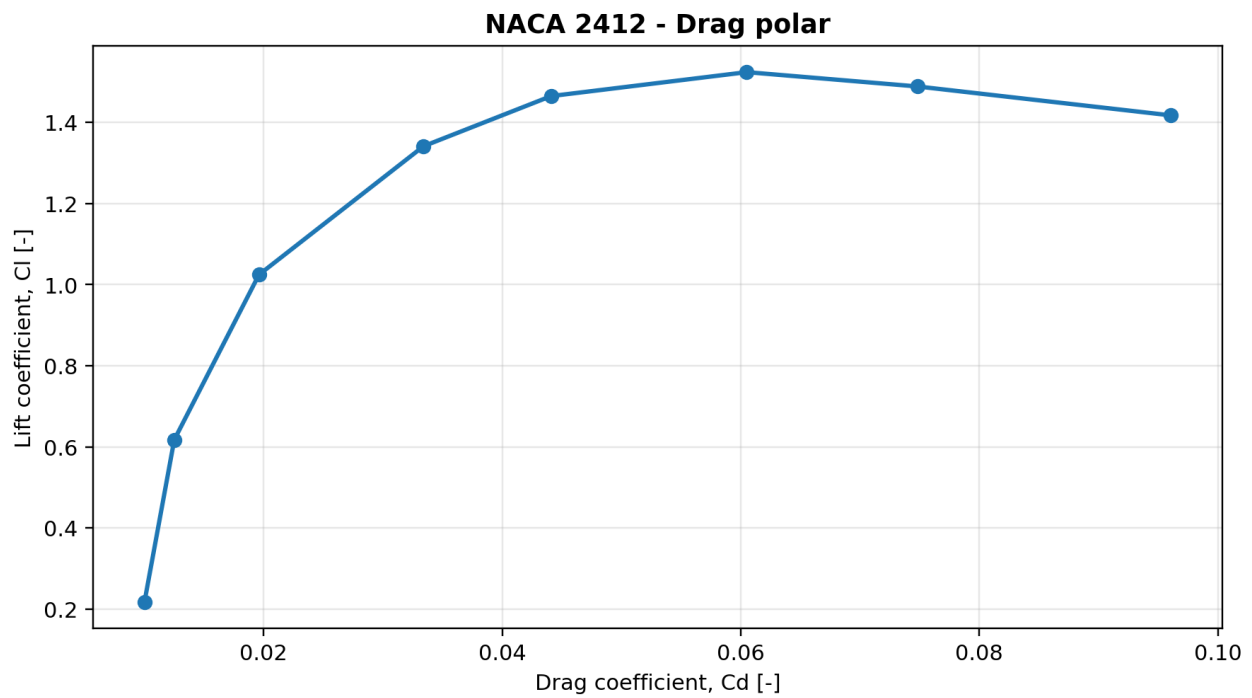


Figure 4 - Computed drag polar for the NACA 2412.

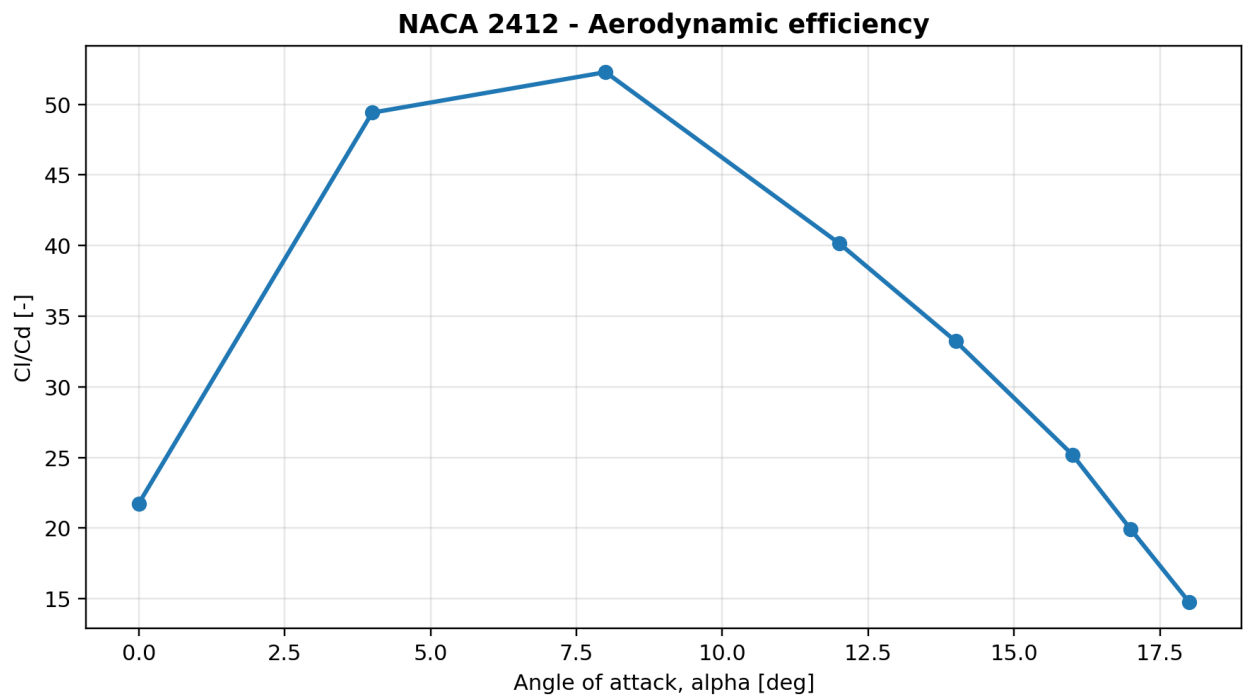


Figure 5 - Aerodynamic efficiency C_l/C_d versus angle of attack.

Within the sampled angles, the best efficiency occurs at 8 degrees, where it is equal to 52.28. This does not prove that the exact optimum is exactly 8 degrees; a finer sweep between 4 and 10 degrees would be required to locate the true maximum. The present result identifies the optimum only within the discrete set of simulated angles.

7. Flow-field evolution and stall onset

The pathlines below are plotted using the same velocity-magnitude range, 0 to 90 m/s, so that color differences can be compared directly. The sequence 0, 8, 16 and 18 degrees illustrates the transition from attached flow to a strongly loaded and increasingly separated state.

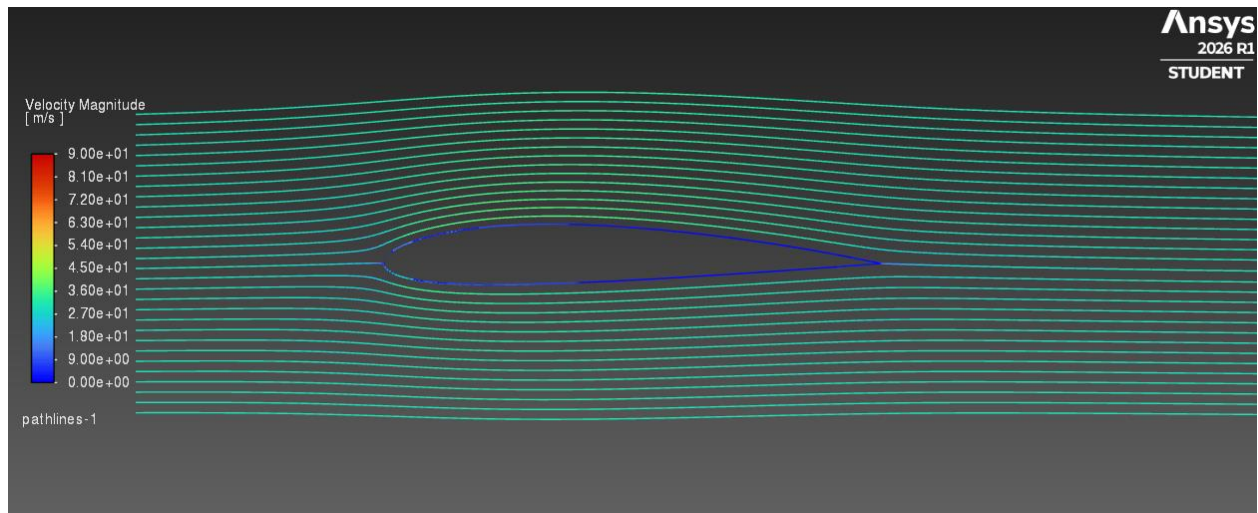


Figure 6 - Velocity pathlines at $\alpha = 0^\circ$ deg. Flow remains essentially attached.

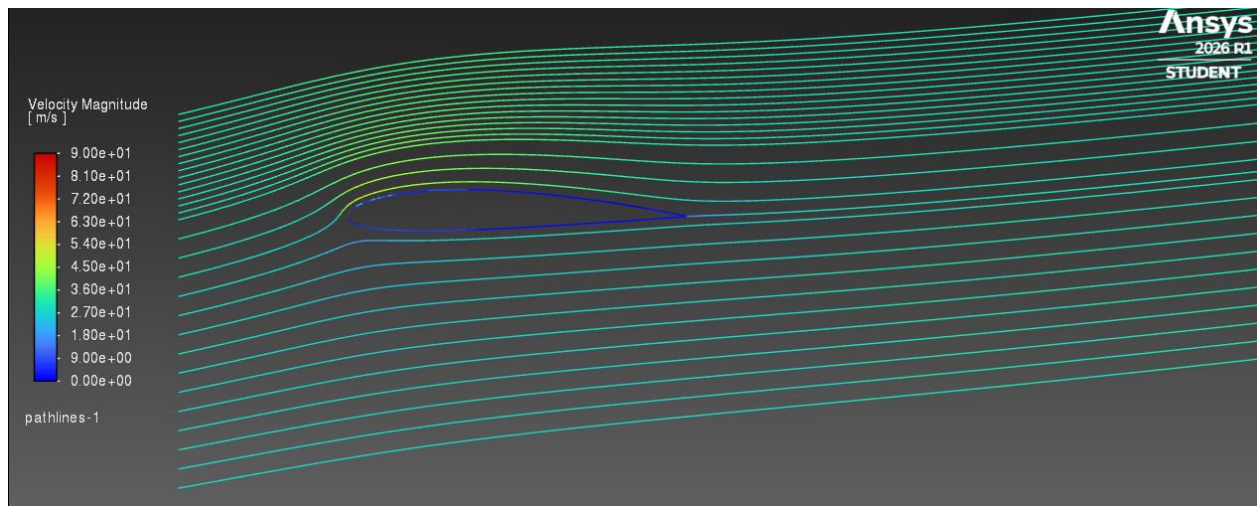


Figure 7 - Velocity pathlines at $\alpha = 8^\circ$ deg. Strong upper-surface acceleration with predominantly attached flow.

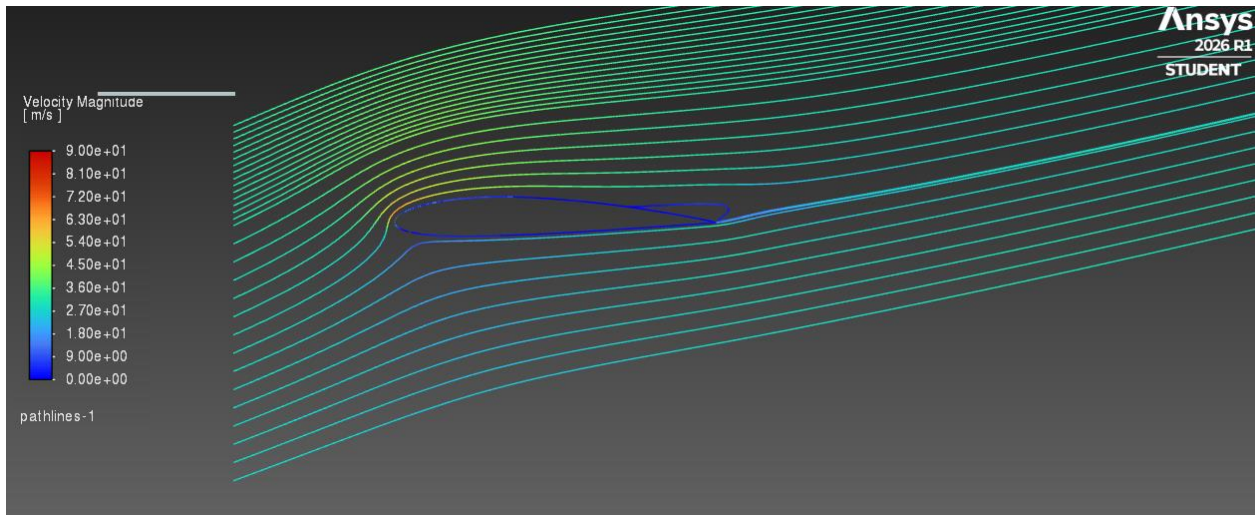


Figure 8 - Velocity pathlines at $\alpha = 16$ deg, corresponding to the highest sampled C_l .

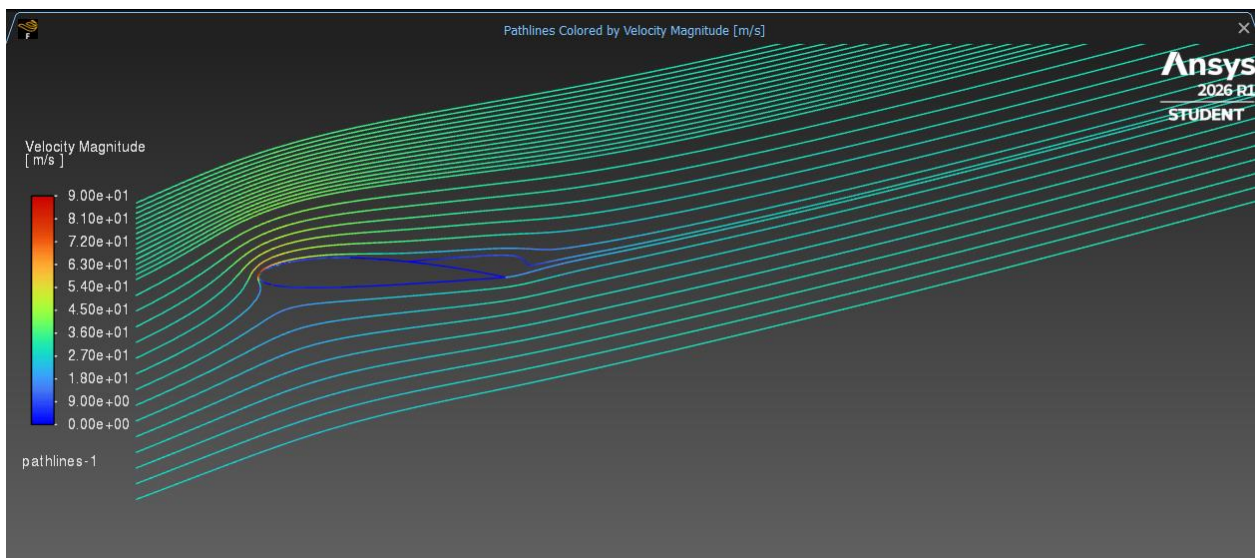


Figure 9 - Velocity pathlines at $\alpha = 18$ deg. The separated low-speed region is more pronounced and lift has decreased.

At zero incidence the cambered profile already accelerates the flow over the upper surface and generates positive lift. At 8 degrees the suction-side acceleration becomes stronger while the flow remains largely attached, explaining the high efficiency. By 16 degrees a low-speed region develops over the aft upper surface and the lift curve has nearly flattened. At 18 degrees the separated region is more evident; simultaneously C_l decreases by about 7% relative to 16 degrees while C_d rises by about 59%. Together, the force trends and pathlines provide consistent evidence that the lift peak lies near 16 degrees for this steady-RANS setup.

8. Limitations and interpretation

The results should be interpreted as a controlled educational CFD study rather than a high-fidelity certification dataset. First, the analysis is two-dimensional and therefore excludes finite wing induced drag and three-dimensional separation. Second, steady RANS can suppress or average unsteady stall dynamics. Third, the sharp closed trailing edge produced a localized poor-quality cell, which can affect robustness and local pressure recovery. Finally, the angle sweep is discrete, so the exact stall angle and exact maximum-efficiency angle are bracketed rather than continuously resolved.

9. Conclusions

The CFD campaign successfully reproduced the expected aerodynamic behavior of a NACA 2412 airfoil at a relatively high Reynolds number. The zero-incidence lift agrees with thin-airfoil theory within approximately 4.6%. The lift curve remains close to linear through 8 degrees, with a measured low-incidence slope of about 0.101 per degree. The highest sampled lift coefficient is 1.5241 at 16 degrees, followed by a reduction at 17 and 18 degrees while drag rises sharply. The best sampled aerodynamic efficiency is 52.28 at 8 degrees.

Most importantly, the force coefficients and flow visualization tell the same physical story: attached and efficient flow at moderate incidence, progressive loss of linearity as the adverse pressure gradient strengthens, and a separated high-drag state after the lift peak. The study therefore provides a coherent baseline for subsequent work on inverted profiles, multi-element aerodynamics and ground-effect configurations.

Appendix A - Key reproducibility settings

Item	Setting
Airfoil	NACA 2412, $c = 1$ m, closed trailing edge
Domain	x : -5c to +11c; y : $\pm 2.5c$
Final mesh sizing	0.15 m global; 0.005 m airfoil edge
Inflation	first layer 2.4e-5 m; 30 layers; growth 1.18
Fluid	$\rho = 1.225$ kg/m ³ ; $\mu = 1.79e-5$ Pa s
Turbulence	k-omega SST; inlet $I = 1\%$; viscosity ratio = 10
Solver	steady pressure-based; SIMPLE; second-order upwind
Reference values	$V_{ref} = 30$ m/s; chord = 1 m; area = 1 m ² per unit span
Sweep	0, 4, 8, 12, 14, 16, 17, 18 deg